

host_agent

Here is a comprehensive summary of everything Ordnance Survey (OS) knows about Exeter, visualized and organized by feature type:

Included Features: - City boundary (polygon) - All buildings (69,169 polygons) - Water features: 265 polygons, 131 points, 135 lines (rivers, lakes, drains, swimming pools, etc.) - Land features: 90,900 polygons, 236 points (arable land, gardens, natural features, etc.) - Landform features: 1,110 polygons, 2,255 lines (cliffs, slopes, mounds, etc.) - Land use: 47,167 polygons (schools, parks, commercial areas, etc.) - Structures: 9 polygons (bridges, tunnels, viaducts, sluices, etc.) - Named addresses and places: 30,981 points (all named locations, organizations, and addresses)

Examples: - Named addresses: LAND OFF, WEAR BARTON ROAD, EXETER, EX2 7EQ; THE SALON, 15, NEWPORT ROAD, EXETER, EX2 7EE; etc. - Building types: Domestic Outbuilding, End-Of-Terrace House, Mid-Terrace House, Unknown Building, etc. - Water features: Watercourse, Drain, Still Water, Swimming Pool, etc. - Land features: Arable Or Grazing Land, Gardens, Scrub, etc. - Structures: Footbridge, Bridge, Pedestrian Tunnel Or Subway, Viaduct, Sluice, etc.

Map Artifact: - Name: exeter_all_features_map - Description: A folium map showing all OS features for Exeter: city boundary, buildings, water, land, landform, land use, structures, and named addresses.

If you want to explore or filter any specific type of feature (e.g., only schools, only rivers, only historic buildings), let me know! I can provide more detail or visualizations for any subset of this data.

plotting_agent

This map visualises all Ordnance Survey features for Exeter, including: - City boundary (1 polygon) - Buildings (69,169 polygons) - Water features: 265 polygons, 131 points, 135 lines - Land features: 90,900 polygons, 236 points - Landform features: 1,110 polygons, 2,255 lines - Land use: 47,167 polygons - Structures: 9 polygons - Named addresses: 30,981 points

First 5 named addresses: - LAND OFF, WEAR BARTON ROAD, EXETER, EX2 7EQ - THE SALON, 15, NEWPORT ROAD, EXETER, EX2 7EE - STREET RECORD, SEABROOK AVENUE, EXETER, EX2 7EF - STREET RECORD, REED WALK, EXETER, EX2 7EF - WATERS MEET, GLASSHOUSE LANE, EXETER, EX2 7BX

First 5 building types: - Domestic Outbuilding - End-Of-Terrace House - Mid-Terrace House - Mid-Terrace House - Unknown Building

First 5 water features: - Watercourse - Drain - Still Water - Still Water - Swimming Pool

First 5 land features: - Arable Or Grazing Land - Arable Or Grazing Land - Arable Or Grazing Land - Arable Or Grazing Land - Arable Or Grazing Land

First 5 structures: - Footbridge - Bridge - Pedestrian Tunnel Or Subway - Viaduct - Sluice

Artifact generated:

- Name: exeter_all_features_map
- Description: A folium map showing all OS features for Exeter: city boundary, buildings, water, land, landform, land use, structures, and named addresses.
- File: exeter_all_features_map.html

Tool Calls and Code

Tool: code executor

```
def plot_exeter_all_features(data: list):  
    import folium  
    import geopandas as gpd  
    from shapely.geometry import mapping  
    import matplotlib  
    import matplotlib.cm as cm  
    import matplotlib.colors as mcolors  
    import random  
  
    # Unpack all datasets  
    (gdf_area, gdf_buildings, gdf_waterpoint, gdf_water, gdf_waterlink, gdf_land, gdf_landpoint,  
     gdf_landform, gdf_landformline, gdf_landuse, gdf_structures, gdf_namedaddr) = data  
  
    # Convert all to EPSG:4326 for folium  
    gdf_area = gdf_area.to_crs(epsg=4326)  
    gdf_buildings = gdf_buildings.to_crs(epsg=4326)  
    gdf_waterpoint = gdf_waterpoint.to_crs(epsg=4326)  
    gdf_water = gdf_water.to_crs(epsg=4326)  
    gdf_waterlink = gdf_waterlink.to_crs(epsg=4326)  
    gdf_land = gdf_land.to_crs(epsg=4326)  
    gdf_landpoint = gdf_landpoint.to_crs(epsg=4326)  
    gdf_landform = gdf_landform.to_crs(epsg=4326)  
    gdf_landformline = gdf_landformline.to_crs(epsg=4326)  
    gdf_landuse = gdf_landuse.to_crs(epsg=4326)  
    gdf_structures = gdf_structures.to_crs(epsg=4326)  
    gdf_namedaddr = gdf_namedaddr.to_crs(epsg=4326)  
  
    # Get centroid for map  
    centroid = gdf_area.geometry.iloc[0].centroid
```

```

m = folium.Map(location=[centroid.y, centroid.x], zoom_start=13, tiles='cartodbpositron')

# Plot Exeter area boundary
folium.GeoJson(gdf_area.geometry.iloc[0], name='Exeter Area', style_function=lambda x: {'color': 'black', 'weight': 3, 'fillOpacity': 0.05}).add_to(m)

# Plot buildings (as polygons, light gray)
folium.GeoJson(gdf_buildings.geometry, name='Buildings', style_function=lambda x: {'color': '#888', 'weight': 0.5, 'fillColor': '#cccccc', 'fillOpacity': 0.1}).add_to(m)

# Plot water polygons (blue)
folium.GeoJson(gdf_waterpoint.geometry, name='Water Polygons', style_function=lambda x: {'color': '#1f78b4', 'weight': 1, 'fillColor': '#1f78b4', 'fillOpacity': 0.7}).add_to(m)

# Plot water points (blue markers)
for idx, row in gdf_water.iterrows():
    folium.CircleMarker(location=[row.geometry.y, row.geometry.x], radius=3, color='#1f78b4', fill=True, fill_opacity=0.7, popup=row.get('description', None)).add_to(m)

# Plot water lines (rivers, canals, etc.)
folium.GeoJson(gdf_waterlink.geometry, name='Water Lines', style_function=lambda x: {'color': '#0077be', 'weight': 2}).add_to(m)

# Plot land polygons (green)
folium.GeoJson(gdf_land.geometry, name='Land Features', style_function=lambda x: {'color': '#228B22', 'weight': 0.5, 'fillColor': '#b2df8a', 'fillOpacity': 0.7}).add_to(m)

# Plot land points (trees, etc.)
for idx, row in gdf_landpoint.iterrows():
    folium.CircleMarker(location=[row.geometry.y, row.geometry.x], radius=2, color='#228B22', fill=True, fill_opacity=0.7, popup=row.get('description', None)).add_to(m)

# Plot landform polygons (brown)
folium.GeoJson(gdf_landform.geometry, name='Landform Features', style_function=lambda x: {'color': '#a0522d', 'weight': 1, 'fillColor': '#deb887', 'fillOpacity': 0.7}).add_to(m)

# Plot landform lines (brown lines)
folium.GeoJson(gdf_landformline.geometry, name='Landform Lines', style_function=lambda x: {'color': '#a0522d', 'weight': 1.5}).add_to(m)

# Plot land use polygons (purple)
folium.GeoJson(gdf_landuse.geometry, name='Land Use', style_function=lambda x: {'color': '#6a3d9a', 'weight': 0.5, 'fillColor': '#cab2d6', 'fillOpacity': 0.7}).add_to(m)

# Plot structures (red)
folium.GeoJson(gdf_structures.geometry, name='Structures', style_function=lambda x: {'color': '#e31a1c', 'weight': 2, 'fillColor': '#fb9a99', 'fillOpacity': 0.7}).add_to(m)

# Plot named addresses (orange points)
for idx, row in gdf_namedaddr.iterrows():
    folium.CircleMarker(location=[row.geometry.y, row.geometry.x], radius=2, color='#ff7f00', fill=True, fill_opacity=0.7, popup=row.get('fulladdress', None)).add_to(m)

folium.LayerControl().add_to(m)
m.save('exeter_all_features_map.html')

# Compose summary
summary = (
    f"This map visualises all Ordnance Survey features for Exeter, including:\n"
    f"- City boundary (1 polygon)\n"

```

```

    f"- Buildings ({len(gdf_buildings)} polygons)\n"
    f"- Water features: {len(gdf_waterpoint)} polygons, {len(gdf_water)} points, {len(gdf_waterlink)} lines\n"
    f"- Land features: {len(gdf_land)} polygons, {len(gdf_landpoint)} points\n"
    f"- Landform features: {len(gdf_landform)} polygons, {len(gdf_landformline)} lines\n"
    f"- Land use: {len(gdf_landuse)} polygons\n"
    f"- Structures: {len(gdf_structures)} polygons\n"
    f"- Named addresses: {len(gdf_namedaddr)} points\n"
    f"\nFirst 5 named addresses: {gdf_namedaddr['fulladdress'].head(5).tolist()}\n"
    f"First 5 building types: {gdf_buildings['description'].head(5).tolist()}\n"
    f"First 5 water features: {gdf_waterpoint['description'].head(5).tolist()}\n"
    f"First 5 land features: {gdf_land['description'].head(5).tolist()}\n"
    f"First 5 structures: {gdf_structures['description'].head(5).tolist()}\n"
)
return [summary, 'exeter_all_features_map', 'A folium map showing all OS features for Exeter: city boundary, buildings, water, land, landform, land use, st
# Do not call the function, just return output as per template.

```

All Tool Calls (in order, except code executor calls which are above)

- send_message: Sent to planning_agent to plan the steps for "Show me everything os knows about Exeter"
- send_message: Sent to named_area to find the area polygon for Exeter
- call_os_ngd: To get area polygons for Exeter
- send_message: Sent to data_analysis_agent to filter the area polygon for Exeter
- generate_metadata_for_artifacts: For 'exeter_area_polygon'
- send_message: Sent to buildings to search for all buildings within the filtered_exeter_area_polygon
- call_os_ngd: To get all buildings in the area
- send_message: Sent to water_features to search for all water features
- call_os_ngd: To get water features
- send_message: Sent to water_network to search for all water network features
- call_os_ngd: To get water network features
- send_message: Sent to data_analysis_agent to filter water network features
- generate_metadata_for_artifacts: For water network artifacts
- send_message: Sent to land_features to search for all land features

- `call_os_ngd`: To get land features
- `send_message`: Sent to `land_use_features` to search for all land use features
- `call_os_ngd`: To get land use features
- `send_message`: Sent to `structures_agent` to search for all structures
- `call_os_ngd`: To get structures
- `send_message`: Sent to `data_analysis_agent` to filter structures
- `generate_metadata_for_artifacts`: For structures
- `send_message`: Sent to `address` to search for all named addresses and places
- `call_os_ngd`: To get named addresses
- `send_message`: Sent to `data_analysis_agent` to filter named addresses
- `generate_metadata_for_artifacts`: For named addresses
- `send_message`: Sent to `plotting_agent` to plot all features
- `generate_metadata_for_artifacts`: For all plotting artifacts

Note

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- All code executor tool calls are shown as code blocks above.
 - All other tool calls and their arguments are listed in sequence.
 - All outputs are shown in the order they occurred.